WORKED SOLUTION

STEP II 2022, Question 5

Pure • Inequalities in a Triangle

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Read the question before you go on. It is in the STEP II paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

The key idea

Splitting the triangle into three pieces from P shows that, however P moves, the weighted sum $ax + by + cz$ stays fixed at 2Δ. Each later part is then an optimisation under that single linear constraint. Part (i) compares $x + y + z$ with the fixed quantity, the identity in (iii)(a) does the same for $x^2 + y^2 + z^2$ (it is the Cauchy–Schwarz inequality in disguise), and for the maximum we use (i) the other way round, with c in place of a.

Solution

Part (i)

Since $a > b$ and $y \geq 0$, we have $by \leq ay$; since $a > c$ and $z \geq 0$, we have $cz \leq az$. Adding ax to both sides of the sum of these gives

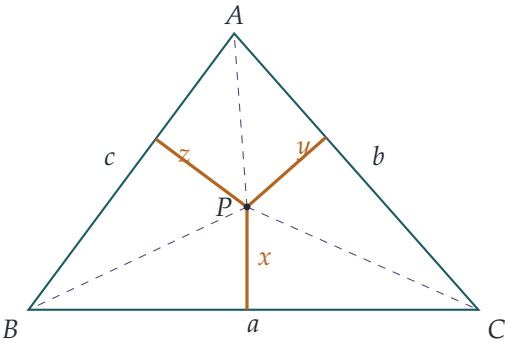
$$ax + by + cz \leq ax + ay + az = a(x + y + z),$$

as required. Because $a - b > 0$ and $a - c > 0$, equality holds exactly when $(a - b)y + (a - c)z = 0$, that is, when $y = z = 0$. We will need this equality case.

Part (ii)(a)

Joining P to the vertices splits ABC into triangles PBC, PCA and PAB (some may be degenerate if P is on a side). Triangle PBC has base $BC = a$ and height x, and so on, so

$$\Delta = \tfrac{1}{2}ax + \tfrac{1}{2}by + \tfrac{1}{2}cz, \qquad \text{that is,} \qquad ax + by + cz = 2\Delta.$$



Conversely, any $x, y, z \geq 0$ with $ax + by + cz = 2\Delta$ are the distances from exactly one point P in the triangle (or on its sides): the three triangle areas $\frac{1}{2}ax$, $\frac{1}{2}by$, $\frac{1}{2}cz$ are non-negative and add up to Δ, and they pin down P (they are its areal coordinates). So in what follows we can treat the problem as purely algebraic: optimise over all non-negative x, y, z with $ax + by + cz = 2\Delta$.

Part (ii)(b)

By part (i) and then (ii)(a),

$$x + y + z \geq \frac{ax + by + cz}{a} = \frac{2\Delta}{a}.$$

Equality holds exactly when $y = z = 0$, and then $ax = 2\Delta$. So the minimum value of the sum is

$$x + y + z = \frac{2\Delta}{a}, \qquad \text{attained at} \quad x = \frac{2\Delta}{a},\ y = 0,\ z = 0.$$

Geometrically, $y = z = 0$ puts P on both CA and AB, so $P = A$, and x is the altitude from A to the longest side BC.

Part (iii)(a)

Expanding the three squared differences,

$$(bx - ay)^2 + (cy - bz)^2 + (az - cx)^2 = b^2x^2 + a^2y^2 + c^2y^2 + b^2z^2 + a^2z^2 + c^2x^2 - 2(abxy + bcyz + cazx),$$

while

$$(ax + by + cz)^2 = a^2x^2 + b^2y^2 + c^2z^2 + 2(abxy + bcyz + cazx).$$

Adding, the cross terms cancel and every product of one of a^2, b^2, c^2 with one of x^2, y^2, z^2 appears exactly once, giving $(a^2 + b^2 + c^2)(x^2 + y^2 + z^2)$. This proves the identity for all real values.

Part (iii)(b)

Put $ax + by + cz = 2\Delta$ into the identity:

$$(a^2 + b^2 + c^2)(x^2 + y^2 + z^2) = (bx - ay)^2 + (cy - bz)^2 + (az - cx)^2 + 4\Delta^2 \geq 4\Delta^2,$$

since squares are non-negative. Hence

$$x^2 + y^2 + z^2 \geq \frac{4\Delta^2}{a^2 + b^2 + c^2}.$$

Ask yourself A lower bound is only the minimum if it is actually reached. Can all three squares vanish at a genuine point P?

Equality needs $bx = ay$, $cy = bz$ and $az = cx$. These say (x, y, z) is a multiple of (a, b, c): if $x = \lambda a$, then $y = \frac{b}{a}x = \lambda b$ and $z = \frac{c}{a}x = \lambda c$, and all three equations hold. The constraint gives $\lambda(a^2 + b^2 + c^2) = 2\Delta$, so $\lambda = \dfrac{2\Delta}{a^2 + b^2 + c^2} > 0$. Then x, y, z are all positive and satisfy $ax + by + cz = 2\Delta$, so by the remark in (ii)(a) they belong to a real point P (strictly inside the triangle). So the minimum value of the sum of squares is

$$\frac{4\Delta^2}{a^2 + b^2 + c^2}, \qquad \text{attained at} \quad x = \frac{2a\Delta}{a^2 + b^2 + c^2},\ y = \frac{2b\Delta}{a^2 + b^2 + c^2},\ z = \frac{2c\Delta}{a^2 + b^2 + c^2}.$$

Part (iv)

Ask yourself Part (i) used that a is the largest side. What happens if we use the smallest side c instead?

Since $a > c$, $b > c$ and $x, y \geq 0$, the argument of part (i) reversed gives $ax + by + cz \geq c(x + y + z)$, with equality exactly when $x = y = 0$. So

$$x + y + z \leq \frac{2\Delta}{c}.$$

Also, as $x, y, z \geq 0$,

$$x^2 + y^2 + z^2 = (x + y + z)^2 - 2(xy + yz + zx) \leq (x + y + z)^2,$$

with equality exactly when $xy + yz + zx = 0$. Putting these together (squaring is safe because both sides are non-negative),

$$x^2 + y^2 + z^2 \leq (x + y + z)^2 \leq \frac{4\Delta^2}{c^2}.$$

Both inequalities become equalities when $x = y = 0$: then $xy + yz + zx = 0$ and $cz = 2\Delta$. So the maximum value of the sum of squares is

$$\frac{4\Delta^2}{c^2}, \qquad \text{attained at} \quad x = 0,\ y = 0,\ z = \frac{2\Delta}{c},$$

which is $P = C$, the vertex opposite the shortest side. (The second inequality is an equality only when $x = y = 0$, so this is the only point where the maximum occurs.)

Checks

Take the acute triangle with sides $a = 7$, $b = 6$, $c = 5$. Heron's formula gives $\Delta = \sqrt{9 \cdot 2 \cdot 3 \cdot 4} = 6\sqrt{6} \approx 14.70$. The answers are then: minimum of $x + y + z$ is $12\sqrt{6}/7 \approx 4.20$; minimum of $x^2 + y^2 + z^2$ is $864/110 = 432/55 \approx 7.85$; maximum is $864/25 = 34.56$. The order makes sense: at $P = C$ the sum of squares is just the altitude from C squared, $(12\sqrt{6}/5)^2 = 34.56$. A computer check over many random acute triangles, with thousands of random points P in each, confirmed $ax + by + cz = 2\Delta$ and all three bounds, with the extremes reached at A, at the point with distances proportional to (a, b, c), and at C.

Common slips

- Giving a lower bound without showing it is attained. In (iii)(b) you must check that the equality point has non-negative distances satisfying the constraint, so that it is a genuine P.
- Mixing up the vertices: the minimum of $x + y + z$ is at A (opposite the longest side), but the maximum of $x^2 + y^2 + z^2$ is at C (opposite the shortest side).
- Dropping the condition $x, y, z \geq 0$ in (iv); without it, $x^2 + y^2 + z^2 \leq (x + y + z)^2$ is false.

Beyond STEP

The identity in (iii)(a) is Lagrange's identity, and it proves the Cauchy–Schwarz inequality $(\mathbf{u} \cdot \mathbf{v})^2 \leq |\mathbf{u}|^2 |\mathbf{v}|^2$ in three dimensions: the minimum point is where (x, y, z) is parallel to (a, b, c), the foot of the perpendicular from the origin to the plane $ax + by + cz = 2\Delta$. The maximum of a convex function such as $x^2 + y^2 + z^2$ over a triangle is always at a corner, a first glimpse of linear and convex optimisation. The point found in (iii)(b) is the symmedian (or Lemoine) point of the triangle. For more on dot products and projections, see Strang, *Introduction to Linear Algebra*.

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